

KUNAL HEMANT PASAD

Machine Learning | NLP | LLM Reasoning & Evaluation

60 Paterson St, New Brunswick, NJ 08901

📞 848-391-7620 ✉ kunalpasad6523@gmail.com 🌐 [Portfolio](#) [LinkedIn](#) [Github](#)

EDUCATION

Rutgers, the State University of New Jersey

Expected May 2027

Master of Science in Computer Science

GPA: 3.83/4.0

Relevant Coursework: Machine Learning, Natural Language Processing, Massive Data Mining, Database Systems

Sardar Patel Institute of Technology, Mumbai

May 2025

Bachelor of Technology in Computer Engineering

GPA: 8.62/10

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL

AI & Machine Learning: PyTorch, scikit-learn, HuggingFace, LLMs, RAG, QLoRA, Embeddings, Prompt Engineering

Backend: FastAPI, Node.js, REST APIs, Distributed Systems

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD, GitHub Actions

Databases: PostgreSQL, MySQL, MongoDB, Redis

Concepts: System Design, Scalable Systems, API Design, Data Pipelines

RESEARCH PUBLICATIONS

Multilingual Sports Question Answering Benchmark

EMNLP 2025

Conference Publication

NLP, LLM Evaluation

- Co-authored multilingual QA benchmark containing over 33K questions across 11 languages and 60 countries
- Designed multi-hop reasoning tasks generating over 30K structured samples for evaluating compositional reasoning capabilities
- Benchmarked LLMs identifying up to 18% performance drop in cross-lingual reasoning across multiple models

IMPULSE: Intelligent Message Prioritization System

Springer ICACIS 2024

Research Publication

Machine Learning, NLP

- Developed message prioritization system using embeddings and gradient boosted models for classification tasks
- Engineered NLP preprocessing pipelines and feature extraction workflows for conversational dataset representation learning
- Achieved 96.55% classification accuracy demonstrating effectiveness of embedding-based prioritization approaches in real-world scenarios

PROFESSIONAL EXPERIENCE

EternIQ

Jan 2025 – May 2025

Machine Learning Intern

Mumbai, India

- Engineered scalable LLM inference pipelines powering AI assistant for reasoning and code generation tasks
- Optimized prompt engineering strategies reducing hallucination rates by approximately 25% across responses
- Designed evaluation framework analyzing 1K model outputs to study reasoning behavior and failure patterns
- Integrated real-time inference services via REST APIs enabling production-grade AI capabilities within systems

PROJECTS

InstantStory – LLM Generation System | Mistral-7B, QLoRA, FastAPI

Mar 2026

- Built end-to-end LLM pipeline generating structured stories from curated dataset of over 13K samples
- Improved narrative coherence by approximately 22% compared to zero-shot baselines using controlled prompting strategies
- Reduced training memory usage by approximately 60% using QLoRA quantization while maintaining generation quality

MindSync – AI Backend Platform | RAG, FastAPI, AWS

Nov 2025

- Developed RAG-based backend system enabling querying unstructured documents with accurate grounded responses
- Evaluated retrieval and prompting strategies reducing hallucinated outputs by 25% across benchmark queries
- Optimized response latency by approximately 30% using caching, efficient chunking, and AWS deployment

Deepfake Guard – ML Pipeline | PyTorch, CNN

Jun 2025

- Developed deep learning pipeline processing over 15K audio samples for robust deepfake detection tasks
- Achieved approximately 93% classification accuracy improving robustness by approximately 18% using augmentation techniques
- Applied explainable AI methods including SHAP and Grad-CAM to analyze model predictions and failure cases